Task 1 — User Information Investigation

Questions:
1. What is your current username?
ak
2. Difference between UID and GID?
UID means user id which identifies the user
GID means group id identifies the users primary group

3. List all groups your user belongs to.
ak adm cdrom sudo dip plugdev users

4. Which group allows administrative access?
adm group allows administrative access

Task 2 — Linux User Database

1. Which file stores Linux user information?
etc/passwd stores the Linux user information

2. Explain ubuntu:x:1000:1000:Ubuntu User:/home/ubuntu:/bin/bash

3. What is the purpose of the login shell field?
login shell field is to tell the system which command-line interface to open for the user when they log in

Task 3 — Linux Group Database

1. Which file stores Linux group information?
/etc/group stores the files of Linux group information

2. What is the purpose of a group in Linux?
to allow a certain permission to certain users.

3. Explain docker:x:999:ubuntu

Task 4 — Permission Analysis
-rw-r--r--.
Owner permissions
the owner can read and write

Group permissions
the group can only read

others permission
the others can only read

Task 5 — Numeric Permission Conversion

Convert rwxrwxrwx, rwxr-xr-x, rw-r--r--, rw-------, r-xr-xr-x to numeric values.

rwxrwxrwx - 777
rwxr-xr-x - 755
 rw-r--r-- - 644
 rw------- - 600
r-xr-xr-x -555

Explain why 777 is dangerous and why 600 is recommended.
in 777 that the file can be accessed by everyone so its danger but in 600 only the user can read and write the file.

Task 7 — Script Execution

Explain execute permission and DevOps usage.
Execute permission is a file system setting that tells the operating system whether a file can be run as a program, script, or process.

Task 8 — Directory Permissions
Create DevOpsLab and analyze drwxr-xr-x.
Explain ls -l vs ls -ld and directory execute permission.
ls -l means listing the files and directory in a lonformat,ls -ld means its also listing in long format but only directories

Task 9 — Ownership Management
Use chown and explain difference between chmod and chown.
chmod is command is used to change the permission of a file ,meanwhile the chown is to change the ownership of the file